Feature-matched masking reveals a dissociation between iconic and working memory

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The authors declare that there are no competing interests.

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Abstract

Visual working memory preserves perceptual information across delay, yet the representational basis of that preservation remains contested. Influential sensory recruitment accounts propose that retention depends on sustained or reinstated perceptual codes in early visual cortex. Under some proposals, this predicts that feature-matched interference delivered to the retinotopic location of a remembered item during delay should disrupt memory-guided report. Across three experiments, we applied the same dynamic, feature-matched masking manipulation across two retention regimes: a 400-ms iconic interval and a multi-second working memory delay.

The results revealed a striking dissociation: masks delivered during a 400 ms iconic window nearly tripled guess rates while leaving precision intact, consistent with disruption of a fragile, sensory-format trace (Experiment 1). In contrast, the identical masks left working-memory performance unchanged at every load, even as load itself reliably impaired performance (Experiment 2). A within-subjects manipulation of mask onset reproduced the dissociation: only masks coincident with sample offset disrupted report, while mask onsets 100 ms or more later left performance intact (Experiment 3).

These results establish a critical constraint on sensory recruitment accounts: once consolidation is complete, multi-second working memory retention does not depend on the sensory-format code that competes with ongoing input for the same feature channels.

Keywords: visual working memory, iconic memory, sensory recruitment, masking, representational format, psychophysics

1. Introduction

Visual working memory supports short-term retention of visual information for comparison, decision, and action. We can compare displays separated by seconds, maintain target features during search, and recover remembered features all while the perceptual system continues to process ongoing input. Any account of visual working memory must therefore explain how perceptual information remains available after stimulus offset during ongoing sensory processing.

1.1. Sensory recruitment and the question of representational format

One influential answer comes from sensory recruitment. On this view, remembered content is maintained through sustained or reinstated perceptual representations in early visual cortex (Harrison & Tong, 2009; Serences et al., 2009; Serences, 2016;

Sreenivasan, Curtis, & D'Esposito, 2014). In nonhuman primates, delay-period activity appears in sensory and sensory-adjacent regions including V4, IT, and MT (Miller & Desimone, 1993; Hayden & Gallant, 2013; Chelazzi et al., 1998; Pasternak & Greenlee, 2005; Bisley et al., 2004; 2001; Zaksas et al., 2001). In humans, remembered features can be decoded from early visual cortex during the delay, and the fidelity of those patterns covaries with psychophysical measures of mnemonic precision (Harrison & Tong, 2009; Serences et al., 2009; Xing et al., 2013; Emrich et al., 2013; Ester et al., 2013). That feature-specific representations in early sensory areas track the content and quality of what is being remembered is precisely why sensory recruitment is compelling: the same code that supports perception appears to carry mnemonic information forward.

In its most basic formulation, sensory recruitment treats working memory maintenance as continued or reinstated perceptual encoding: the system bridges the retention interval by keeping the original sensory code active in the same feature-selective, retinotopically organized populations that encoded it (Serences, 2016; Sreenivasan, Curtis, & D'Esposito, 2014). Such channel-specificity also carries a direct empirical prediction: e.g., if remembered orientations are maintained in the same feature-selective, retinotopically organized populations that encoded them, then subsequent input that drives those same populations during the delay should perturb the mnemonic signal.

Yet whether working-memory report *depends* on the continued integrity of that sensory-format code has proven difficult to establish. The evidence for sensory recruitment comes from decoding: feature-specific delay-period activity is recoverable in early visual cortex, and recovery precision tracks behavioral performance (Harrison & Tong, 2009; Emrich et al., 2013; Ester et al., 2013). But this evidence is correlational: the presence of decodable feature information in a region does not by itself establish that the region's activity constitutes the substrate on which report depends (Riggall & Postle, 2012; Xu, 2017). A region may carry feature information as a byproduct of feedback from higher areas (e.g., Van Kerkoerle et al., 2017; Lawrence et al., 2018) as a readout-preparation signal that anticipates use rather than constituting the basis for storage (e.g., Stokes, 2015; Wolff et al., 2017; Myers, Stokes, & Nobre 2017), or as an attentional or priority trace coupled to an item’s location rather than a content-specific mnemonic code (e.g., Bisley & Goldberg, 2010; Bettencourt & Xu, 2016)- each of which would be recoverable by decoding without being causally necessary for retention.

This distinction between presence and dependence constitutes a central tension in the field, and these dissociations between neural decodability and behavioral performance have motivated accounts in which working memory does not maintain a simple copy of

the sensory code. Such accounts come with a variety of commitments. One prevalent model suggests that representations may be transformed within visual cortex into orthogonalized formats organized around task demands rather than the retinotopic and feature-specific structure of the original sensory encoding, possibly via rotation, compression, or categorical recoding (Murray et al., 2017; Kiyonaga & Serences, 2025). They may coexist with ongoing sensory encoding within overlapping neural populations but occupy separable subspaces that insulate mnemonic from sensory content (Rademaker, Chunharas, & Serences, 2019).

Other accounts place maintained representations outside the feedforward sensory pathway altogether: storage may enlist top-down feedback signals (e.g., from fronto-parietal regions) to early visual cortex that target feedback-recipient layers rather than feedforward-recipient layers (van Kerkoerle et al., 2017; Lawrence et al., 2018) or in activity-silent synaptic states that carry no sustained firing and are therefore invisible to conventional decoding (Stokes, 2015; Wolff et al., 2017).

A representation orthogonalized within V1 but still anchored to populations driven by the original feedforward pathway would remain susceptible to interference targeting that pathway; a representation maintained by feedback signals or by activity-silent synaptic states would not. Adjudicating among them therefore requires testing whether maintenance *depends* on a sensory representation that remains coupled to the same feature channels that support initial encoding.

What is central for the current investigation is whether a durable store of the visual stimulus remains available even after disruption of iconic memory and the sensory representations that subserve the feedforward sweep.

1.2. The Iconic store and sensory-format dependence

The question of sensory-format dependence already has a clear answer in one retention regime. A rich literature characterizes *iconic memory* as a brief, high-capacity sensory store that retains the retinotopic organization and low-level feature selectivity of the original sensory encoding (Bradley & Pearson, 2012; Coltheart, 1980; Sperling, 1960). We expect that iconic representations remain embedded in the same sensory machinery that processes incoming input, as such, they remain vulnerable to overwrite from subsequent stimulation that recruits overlapping feature channels (Averbach & Coriell, 1961; Breitmeyer & Öğmen, 2006; Di Lollo, 1980). The vulnerability of this sensory substrate has been characterized extensively in visual masking paradigms, with especially strong disruption when masks share feature content and retinotopic position with targets (Averbach & Coriell, 1961; Breitmeyer & Öğmen, 2006; Coltheart, 1980; Di Lollo, 1980; Sperling, 1960). Of course, iconic memory and visual working memory are not equivalent: they differ in capacity, time-course, and putative locus (Sperling 1960,

Phillips, 1974), but the most straightforward sensory recruitment accounts commit them to a shared representational substrate: feedforward-coupled feature channels in early visual cortex (Harrison & Tong, 2009; Serences et al., 2009; Ester et al., 2013). That is, to the extent that residual activity in V1 is dependent on the feedforward signal for maintenance, the supported representations will be vulnerable to masking.

Feature-matched masks therefore engage precisely the substrate that sensory recruitment accounts identify as the working memory store. This makes iconic memory a uniquely useful positive control for sensory-format dependence. To the extent that any storage mechanism depends on the rapid feedforward coding of early visual features—whether via an immediate sensory representation, sustained attention to those features, or rapid recoding within early visual cortex—it may inherit iconic vulnerability to feature-matched interference. A null feature-matched masking effect on working memory under conditions that successfully disrupt iconic memory would then implicate a representational substrate insulated from bottom-up sensory drive: that is, protected from the very interference that overwrites the iconic store.

Existing evidence has yet to resolve this prediction cleanly. Indeed, a variety of masking studies have examined the effects of sensory interference on working memory retention, and report impairment (Phillips, 1974; Sreenivasan & Jha, 2007; Vogel et al., 2006; Xu, 2017), but at least three limitations complicate interpretation. First, interference typically occurred only during early consolidation; once consolidated, working memory was unaffected by masking, leaving open whether maintenance-period representations are truly robust or whether the timing of interference simply missed the vulnerable window. Second, in the few cases where null interference effects on working memory were reported, no sensory baseline was established to verify mask effectiveness, making it impossible to distinguish genuine working memory robustness from an ineffective manipulation (Sreenivasan & Jha, 2007). Third, masking strength depends critically on target-mask feature correspondence (Rademaker et al., 2015): masks that share spatial frequency, orientation bandwidth, and retinotopic position with remembered items produce stronger disruption precisely because they recruit overlapping sensory channels (Bruchmann et al., 2010; Drewes et al., 2023; White & Lorber, 1976). Without establishing that a masking manipulation successfully engages sensory-format representations under matched conditions, a null working-memory outcome carries no clear interpretation.

1.3. The present experiments

To ask whether working memory depends on the same sensory-format representation as iconic memory, we applied the same dynamic, feature-matched masking manipulation across two retention regimes. Experiment 1 characterized the effect of backward masking on retention across a 400-ms iconic interval. Experiment 2 applied the identical manipulation across a five-second working memory delay, crossing mask condition with memory load (1–3 items). Experiment 3 then reproduced this dissociation within a single iconic-scale paradigm, varying only mask onset time.

If working memory storage relies on the same sensory-format trace as iconic memory, masks that disrupt iconic availability should produce the same vulnerability signature during the working memory delay. Masking selectively disrupted iconic memory while leaving working memory performance unchanged. The transition from sensory persistence to working memory maintenance therefore requires a change in representational format to an insulated, durable store.

2. General methods

2.1 Participants

Each experiment recruited 40 college-age students from the Johns Hopkins University SONA Subject Pool to participate in exchange for course credit. All subjects were over the age of 18 and reported normal or corrected-to-normal vision. All procedures were approved by the Institutional Review Board at Johns Hopkins University in accordance with the Declaration of Helsinki. All participants were naive to the experimental hypotheses and provided informed consent prior to participation.

2.2 Apparatus and setup

Stimuli were generated and presented using custom PsychoPy code on a BenQ Zowie desktop computer projecting to a 1920×1080px display with a 100 Hz refresh rate. Centrally-fixating observers (enforced via desktop-mounted Eyelink 1000 eyetracker; SR Research) were tested in a dimly lit, sound-attenuated room, maintained with a forehead and chinrest.

2.3 Stimuli

Stimuli were sine-wave gratings presented at fixed retinotopic locations around a central fixation cross. Two mirrored sets of three positions were used; within each set, locations were equidistant from one another and from fixation. Stimulus centers were positioned 5.14° from fixation. Sample gratings had a radius of 1.285°, phase of 0, contrast of 1, and spatial frequency of 5 cycles per stimulus (scaled with stimulus size in PsychoPy).

Mask gratings had a radius of 1.928° and matched the sample gratings in spatial frequency and contrast. A dashed probe ring (radius = 2.571°) was used in attention-check probe trials. On mask trials, target locations were flooded with rapidly changing, high-contrast gratings that shared spatial frequency, contrast, and retinotopic position with the sample. Mask orientations were sampled uniformly across orientation space with the constraint that successive orientations differed by at least 30°.

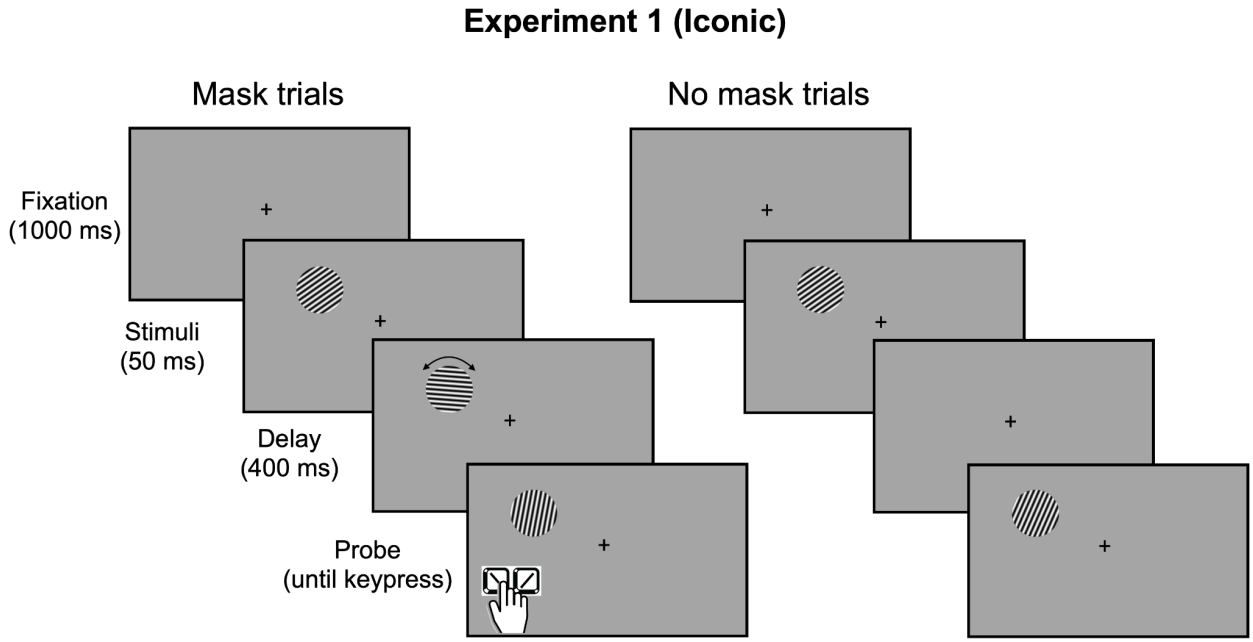

Figure 1. Sample task and trial sequence from Experiment 1 (Iconic task). Each trial began with central fixation. A single peripheral grating appeared briefly (50 ms) at one of the predefined stimulus locations, followed by a 400 ms retention interval. During the interval, the trial was either unmasked (blank screen) or masked, in which the previously occupied location was filled with rapidly changing, feature-matched mask gratings (orientation updated every 50 ms). At test, a single grating appeared at the same location, randomly rotated relative to the sample by ±10°, ±20°, ±30°, or ±40°. Observers reported whether the orientation change was clockwise or counterclockwise.

3. Experiment 1

Experiment 1 tested whether dynamic, feature-matched masks impair performance in a purely iconic memory task. If performance in this regime depends on a sensory-format trace, then masks that recruit the same orientation channels at the same spatial positions should disrupt usable information.

3.1 Experiment 1 Methods

Each trial began with a central fixation cross. A single, randomly oriented grating appeared for 50 ms at one of six predefined spatial locations (**Figure 1**). The retention interval lasted 400 ms. On no-mask trials, the screen remained blank. On mask trials, the previously occupied location was flooded with rapidly changing mask gratings for the duration of the interval (orientation updated every 50 ms). These timings were chosen to optimize performance with respect to the time-course of iconic memory (e.g., Coltheart, 1983; Irwin and Yeomans, 1986). At test, a single grating appeared in the same location, rotated relative to the sample by ±10°, ±20°, ±30°, or ±40°. Participants indicated whether the change was clockwise or counterclockwise relative to the encoded grating at that location. Each participant completed 80 trials. Mask condition (mask vs. no mask) was manipulated within subjects.

3.2 Experiment 1 Results

Performance was modeled for each participant using a cumulative normal psychometric function (psignifit 4; Schütt et al., 2016), fit separately for mask and no-mask conditions. The psychometric function relates the signed orientation difference between test and sample (in degrees) to the probability of a "clockwise" response (**Figure 2A**). From each fit, we extracted two parameters: precision ($1/\sigma$), indexing the fidelity of the orientation representation, and guess rate (γ), indexing the lower and upper asymptote, corresponding to the probability that orientation information was entirely unavailable at test. Twelve participants were excluded due to at least one invalid psychometric fit (**Supplementary S1**), leaving n = 28 in subsequent analyses.

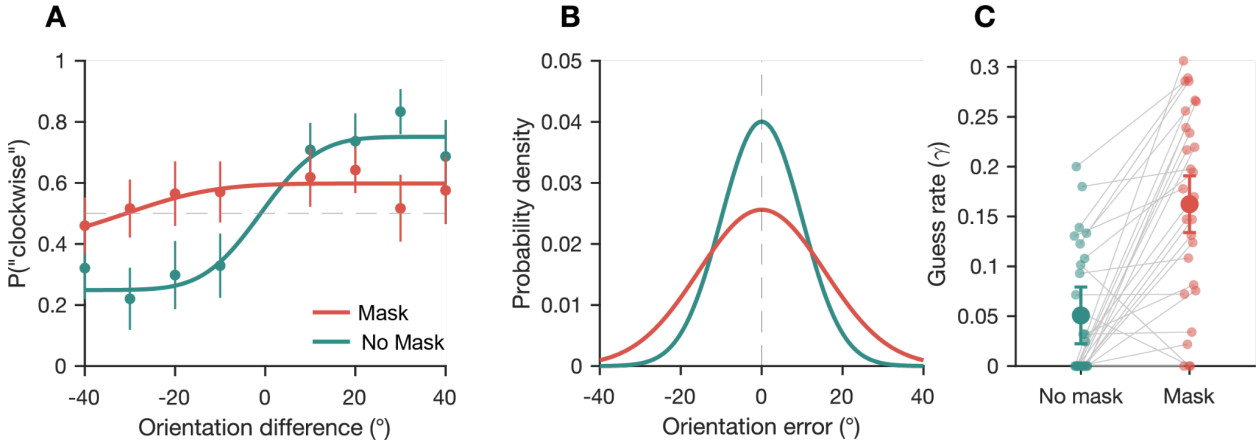

Figure 2: Masking selectively increases guess rate during iconic retention. (A) Group psychometric functions for the masked (orange) and unmasked (teal) conditions in Experiment 1. Points show the mean proportion of "clockwise" responses across participants at each signed orientation difference (test minus sample, degrees); solid curves are cumulative Gaussian fits. Error bars denote 95% between-subject

confidence intervals. **(B)** Precision depicted as Gaussian probability density functions. Curves depict the internal noise distribution implied by fitted sigmas from group-level psychometric fits in **(A)**. **(C)** Guess rate (gamma). Individual participant data (small dots); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

3.2.1 Precision

Precision estimates in the iconic task did not differ reliably between mask and no-mask trials ($t(27) = 1.763$, $p > .05$, $BF_{10} = 0.87$; **Figure 2B**). Although the mean precision estimate was numerically higher in the mask condition ($M = 0.345$) than in the no-mask condition ($M = 0.148$), the Bayes factor indicated anecdotal-to-no evidence for this effect. Thus, when orientation information was successfully retained, its fidelity was not degraded by masking.

3.2.2 Guess rate

In contrast, masking robustly increased guess rates. The γ parameter was more than three times higher in mask trials ($M = 0.162$) compared to no-mask trials ($M = 0.051$), $t(27) = 5.686$, $p < .001$, $BF_{10} \approx 11575$. This masking effect was expressed as a reduction in the asymptote of the psychometric function, consistent with a total loss of usable sensory information, rather than degraded precision (**Figure 2C**). This result constitutes decisive evidence that masking increased the probability that orientation information was unavailable at test for iconic memory.

3.2.3 Effective precision

Because masking substantially elevates guess rates, raw precision estimates ($1/\sigma$) may be inflated through a known slope-asymptote trade-off in cumulative Gaussian models (Wichmann & Hill, 2001; Prins, 2012). We therefore computed an effective precision measure that scales $1/\sigma$ by the proportion of trials containing usable information (see **Supplementary S2**). Effective precision did not differ between mask and no-mask trials ($t(27) = 0.775$, $p = .445$, $BF_{10} = 0.26$), confirming that masking reduces iconic availability without altering the fidelity of the surviving iconic representation.

4. Experiment 2 (VWM)

Experiment 2 applied an identical masking manipulation across a multi-second delay to ask whether the same feature-matched interference that eliminated iconic availability

would similarly impair working memory performance in an otherwise matched task. The paradigm and stimuli matched Experiment 1, extended across a multi-second delay (5 s) and longer encoding duration (1.2 s) consistent with the known time-course of visual working memory. We additionally crossed mask condition with memory load (1–3 items) to test a specific prediction of sensory recruitment: if maintenance occupies the same feature-selective populations that encoded the samples, with per-item fidelity degrading as those populations are divided across more items (e.g., Emrich et al., 2013), then feature-matched interference should scale with load.

4.1 Experiment 2 Methods

The spatial layout, stimulus properties, and mask characteristics remained identical to Experiment 1. Mask condition (mask vs. no mask) and memory load (1-3 items) were manipulated within subjects. 24 probe trials were randomly interspersed to ensure sustained attention during the delay. On probe trials, a dashed ring appeared 2-4 seconds after sample offset for 100 ms at one of the sample locations, and participants pressed a key to report its presence.

4.2 Experiment 2 Results

The same psychometric modeling procedure was applied at each level of load (1, 2, or 3 items) and mask condition (**Figure 3A**). Twelve participants were excluded due to inattention that resulted in at least one invalid psychometric fit across the six conditions (3 loads X 2 mask conditions), leaving N = 29. Because Experiment 2 crossed two within-subjects factors (mask and load) we analyzed each dependent variable using a 2 X 3 repeated-measures ANOVA alongside Bayesian model comparisons (BIC-based Bayes factors from nested linear mixed-effects models).

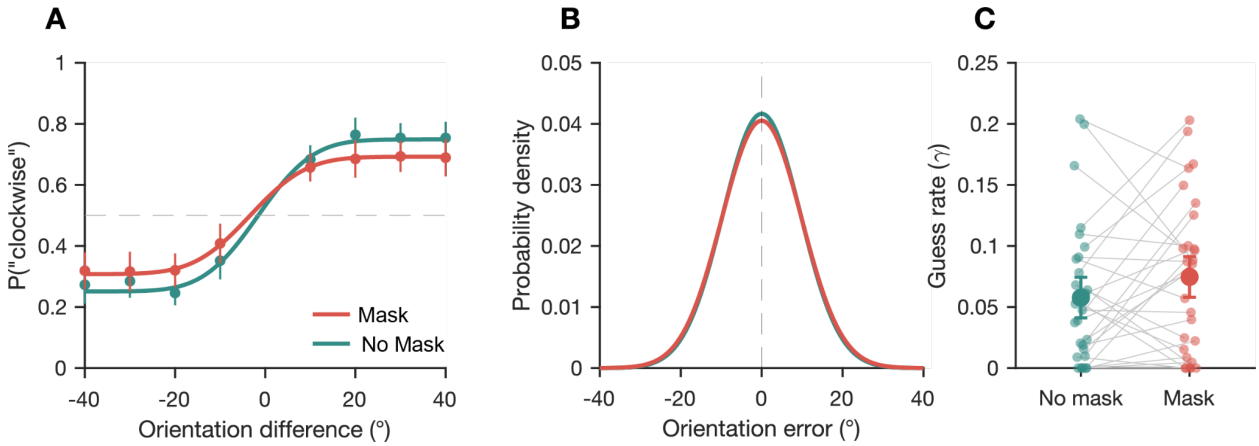

Figure 3. Delay-period masking leaves working memory performance intact. (A) Group psychometric functions (collapsed across load) for the masked (orange) and

unmasked (teal) conditions in Experiment 2. **(B)** Precision depicted as Gaussian probability density functions. Curves depict the internal noise distribution implied by fitted sigmas from group-level psychometric fits in **(A)**. **(C)** Guess rate (gamma). Individual participant data (small dots); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

4.2.1 Precision

A 2 (mask) X 3 (load) repeated-measures ANOVA on precision estimates revealed no significant main effect of load, $F(2, 56) = 0.794$, $p = .457$, no main effect of mask, $F(1, 28) = 1.409$, $p = .245$, and no load X mask interaction, $F(2, 56) = 0.181$, $p = .835$ (**Figure 4A**). Bayesian model comparisons converged (BF_{10} for mask = 0.133, BF_{10} for load = 0.014, BF_{10} for interaction = 0.007), providing substantial-to-strong evidence favoring the null. Mean precision declined numerically with increasing load (Load 1: M = 0.274; Load 2: M = 0.214; Load 3: M = 0.176), but this trend was not statistically reliable. Collapsed across load, mask and no-mask trials did not differ in precision, $t(28) = -1.187$, $p = .245$, $BF_{10} = 0.38$ (**Figure 3B**).

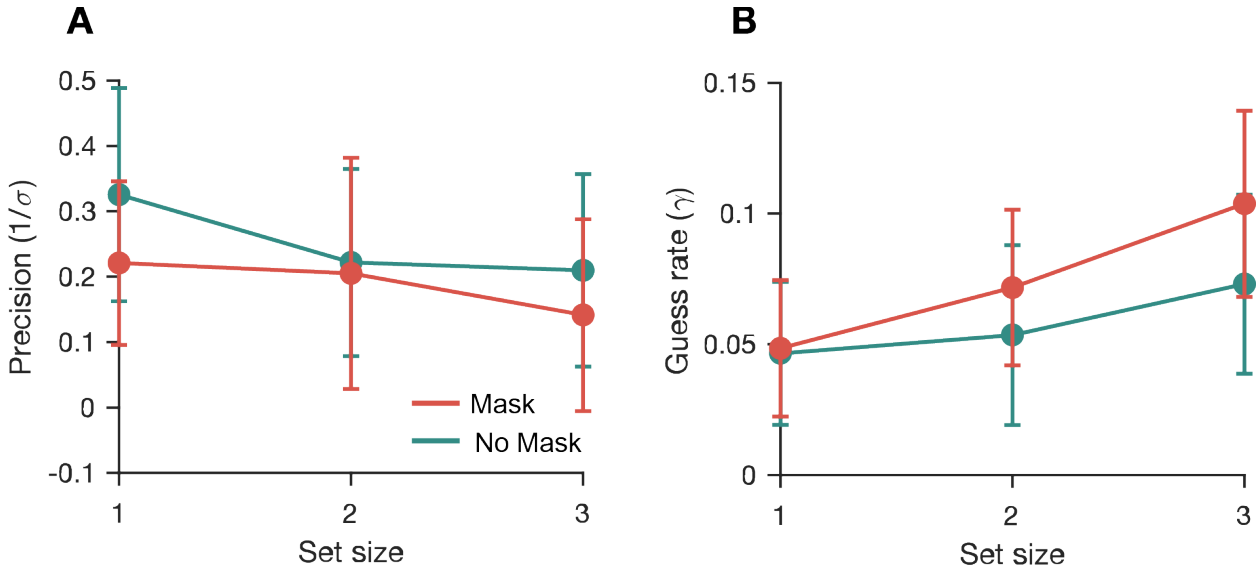

Figure 4. Load-dependent degradation without masking effects in visual working memory. (A) Model-derived precision (1/σ) estimates as a function of item load (1, 2, 3), shown separately for the no-mask (teal) and mask (orange) condition. Precision declines systematically with increasing load, and does so independently of masking condition. There was no main effect of masking. **(B)** Model-derived guess rate (γ) estimates across load, plotted separately for no-mask and mask trials. Guessing increases with load, consistent with reduced availability at higher set sizes. Masking does not reliably alter guess rate at any load, and no load X mask interaction was

observed. Error bars denote within-subject 95% Cosineau-Morey CIs across the full 2x3 design.

4.2.2 Guess rate

A 2 X 3 ANOVA on guess rate revealed a significant main effect of load, $F(2, 56) = 3.877$, $p < .05$, $BF_{10} = 0.208$, reflecting increasing guess rates at higher set sizes (**Figure 4B**). Critically, there was no main effect of mask, $F(1, 28) = 2.165$, $p > .1$, $BF_{10} = 0.193$, and no load X mask interaction, $F(2, 56) = 0.385$, $p > .5$, $BF_{10} = 0.009$. Collapsed across load, mask ($M = 0.075$) and no-mask ($M = 0.058$) trials did not differ in guess rate, $t(28) = 1.471$, $p > .1$, $BF_{10} = 0.55$ (**Figure 3C**).

4.2.3 Effective precision

As in Experiment 1, an effective precision measure (see **Supplementary S2**) was computed for each participant at each load and mask condition. A 2 X 3 ANOVA revealed a significant main effect of load ($F(2, 56) = 3.483$, $p = .038$) but no main effect of mask and no load X mask interaction (both $BF_{10} < 0.18$). Collapsed across load, effective precision did not differ between no-mask ($M = 0.169$) and mask ($M = 0.123$) conditions ($t(28) = -1.615$, $p > .1$, $BF_{10} = 0.68$).

5. Experiment 3

Experiment 3 held the encoding regime, retention interval, task, and mask properties of Experiment 1 constant, and varied only mask onset time. This isolated mask onset as the sole determinant of whether interference coincides with the iconic-vulnerable window.

5.1 Experiment 3 Methods

Each trial consisted of a 50 ms sample grating followed by a fixed 400 ms retention interval. In the mask condition, mask duration varied across five levels (50 ms, 100 ms, 200 ms, 300 ms, 400 ms). Because mask offset was anchored to probe onset, mask onset time varied as a function of duration (onset = 400 - duration). Thus, longer durations resulted in earlier onset and more sustained occupation of the same feature-

selective channels during retention.

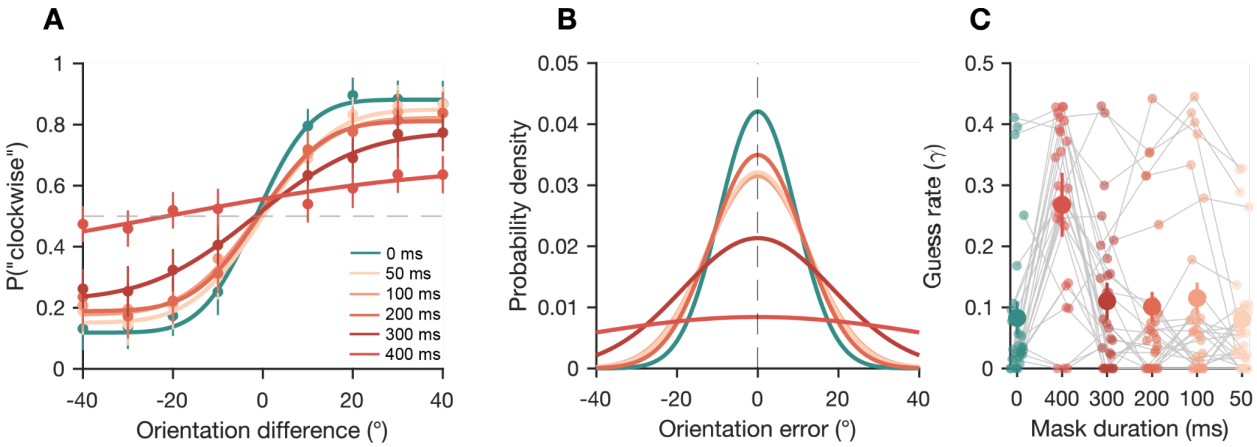

Figure 5. Masking impairment depends on mask onset. (A) Group psychometric functions for the unmasked condition (teal) and each of five mask durations (peach-to-coral gradient, light = 50 ms to dark = 400 ms) in Experiment 3. Points show the mean proportion of "clockwise" responses across participants at each signed orientation difference (test minus sample, degrees); solid curves are cumulative Gaussian fits. Error bars denote 95% between-subject confidence intervals. **(B)** Precision depicted as Gaussian probability density functions. Curves depict the internal noise distribution implied by fitted sigmas from group-level psychometric fits in (A). **(C)** Guess rate (gamma). Individual participant data (small dots, connected across conditions by thin gray lines); group means with within-subject 95% Cousineau-Morey confidence intervals (large dots with error bars).

5.2 Experiment 3 Results

Collapsed across mask duration, precision did not differ between the mask and no-mask condition (M = 0.209 vs. 0.312), $t(24) = -0.97$, $p = .344$, $BF_{10} = 0.32$. In contrast, masking reliably increased guess rate, $t(24) = 3.52$, $p = .002$, $BF_{10} = 36.05$. To characterize the temporal profile, we ran a one-way repeated-measures ANOVA on guess rate across all six conditions (no-mask + five mask durations), which revealed a reliable condition effect, $F(5, 120) = 17.25$, $p < .001$. Follow-up Bonferroni-corrected pairwise comparisons against the no-mask condition revealed a difference only in the 400 ms masking condition, $t(24) = 6.17$, $p < .0001$, $BF_{10} = 28{,}787.31$ (50 ms: $BF_{10} = 0.20$; 100 ms: $BF_{10} = 0.61$; 200 ms: $BF_{10} = 0.62$; 300 ms: $BF_{10} = 0.49$; all p's > .1).

Because a linear duration trend on guess rate was statistically reliable in the full dataset (slope = 0.0004 per ms, $t(123) = 6.32$, $p < .001$), we ran a sensitivity analysis excluding the 400 ms condition; the trend did not survive (slope = 0.0001/ms, $t(98) = 1.40$, $p > .1$), confirming that this trend was driven entirely by the 400 ms condition. That is, only the

400 ms mask (the single condition in which mask onset coincided with sample offset) impaired report. Masks beginning 100 ms or more after sample offset, even when they occupied the same feature channels for most of the delay period, had no measurable effect on performance.

6. Discussion

The present experiments asked a direct question about representational format: does visual working memory storage depend on the same sensory-format code that supports iconic persistence? The answer, across all measures and all load levels, was no.

In Experiment 1, dynamic, feature-matched masks delivered during a 400 ms iconic retention interval nearly tripled guess rates relative to unmasked trials, while leaving the precision of surviving representations intact. Under masking, participants in the iconic condition behaved as though no usable orientation information remained: a profile consistent with wholesale disruption of a fragile sensory-format trace. In contrast, working memory performance showed no such signature. Identical, feature-matched masks, sustained throughout a five-second working memory delay, neither raised the probability of information loss nor altered the precision of retained representations at any memory load. A direct between-experiment comparison yielded a pronounced dissociation in mask-induced guessing (**Supplementary S5**); that is, the same masking condition that disrupted iconic persistence left working memory intact. Experiment 3 reproduced this dissociation within a single iconic-scale paradigm: the mask condition matching Experiment 1 replicated its disruption, while mask onsets just 100 ms later replicated the null effects observed in Experiment 2.

6.1 The behavioral signature of sensory-format maintenance

Experiment 1 established the signature of sensory-format vulnerability in this paradigm. Feature-matched masks reliably disrupted the availability of iconic representations, providing direct evidence that the masking manipulation engages the relevant sensory channels with sufficient feature overlap, at the right retinotopic locations, during maintenance itself (Sreenivasan & Jha, 2007; Bruchmann et al., 2010; Rademaker et al., 2015). Experiment 2 tested whether vulnerability to masking would scale with working memory demands by crossing mask condition with memory load (1–3 items). Expected load effects were observed: precision declined and guessing increased as set size increased from one to three items, confirming that the paradigm was sensitive to capacity constraints. Critically, neither precision nor guessing was affected by mask presence at any load. Even when working memory neared capacity, the stored representation remained insulated from the same sensory input that eliminated iconic

availability. This interaction (load sensitivity without mask sensitivity) is difficult to accommodate under any account in which working memory storage remains coupled to the same sensory channels targeted by incoming sensory input.

6.2 The transition from sensory-format dependence is fast and sharp

Experiments 1 and 2 differ in encoding duration and retention interval; Experiment 3 held both parameters constant, varying only mask onset, and replicated the same dissociation. Within subjects and within the same 400 ms delay, mask onsets timed to sample offset replicated Experiment 1's disruption, while masks onset 100 ms or later had no measurable effect on performance. That is, a 300 ms mask occupying the same feature channels and retinotopic locations for three-quarters of the retention interval produced no measurable cost, while a 400 ms mask in the same paradigm eliminated usable information on a substantial number of trials. If a single, sensory-format representation were progressively degraded by ongoing interference, longer masks should produce larger costs; they did not. The window within which report depends on the interference-vulnerable substrate is therefore narrow, with a sharp, rather than graded transition. This profile implicates a handoff between representational formats, with the sensory-format trace giving way to a code that tolerates continued drive of the same channels.

6.3 A boundary condition on accounts of perceptual reuse

This dissociation constitutes direct behavioral evidence against any sensory recruitment account in which working memory maintenance depends on the continued integrity of a stimulus-coupled perceptual code occupying the same feature channels and retinotopic positions as the remembered target (Serences, 2016; Sreenivasan et al., 2014).

While these results establish a critical boundary condition on sensory-based storage accounts, several contemporary alternatives are consistent with a representational format that tolerates ongoing feature-matched interference. For instance, the perceptual code established during encoding may be recoded into an orthogonal format that reduces representational overlap with the incoming sensory stream (Murray et al., 2017; Rademaker, Chunharas, & Serences, 2019), consistent with rotational dynamics that separate sensory from mnemonic codes within overlapping populations (Libby & Buschman, 2021). Alternatively, working memory contents may be stored within activity-silent synaptic states (Stokes, 2015; Wolff et al., 2017), or distributed frontoparietal circuits, with delay-period activity in early visual cortex reflecting feedback rather than storage (Bettencourt & Xu, 2016; Postle, 2006).

These results are also consistent with recent evidence that working memory representations in parietal cortex are maintained in a recoded, abstract format that retains task-relevant feature information decoupled from the stimulus-evoked code (Duan & Curtis, 2024). The behavioral dissociation we report is consistent with this kind of abstraction: a maintained representation no longer coupled to the same feature channels would preserve report while tolerating feature-matched interference at the remembered location.

6.4 Behavioral dependence as a complement to decoding evidence

The present findings bear on a recurrent methodological tension in this literature. Much of the evidence for sensory recruitment comes from multivariate decoding: feature-specific delay-period activity can be read out from early visual cortex, and the fidelity of that readout covaries with behavioral precision (Harrison & Tong, 2009; Serences et al., 2009; Emrich et al., 2013; Ester et al., 2013). Importantly, decodability does not establish dependency: a cortical region may carry feature information (e.g., reflecting feedback, attentional modulation, or readout preparation) without that information constituting the representational substrate that report depends on (Riggall & Postle, 2012; Xu, 2017).

Here, we provide a complementary behavioral approach by first demonstrating that a given interference stream reliably disrupts the sensory-format substrate, then asking whether a second regime shares that vulnerability. Working memory's insulation from the interference that eliminated iconic availability is evidence that maintenance does not depend on the same retinotopically organized feature-selective channels that support iconic retention.

6.5 Conclusion

Sensory recruitment captures a genuine organizational principle of the visual system: early visual cortex provides the kind of high-dimensional, feature-selective code that is well-suited to representing perceptual content. The present results establish a boundary on how far this principle extends: once consolidation is complete, multi-second retention does not depend on the continued integrity of a sensory-format trace that shares the channels and vulnerability profile of iconic memory.

Iconic persistence and working memory maintenance are therefore distinguished not only by distinct time courses, but by the representational format and interference profile that characterize each. Any successful account of visual working memory must therefore contend with the representational discontinuity between perception and

memory: the feature-selective, retinotopically-organized format that supports perception is not the code that supports memory across delay.

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SUPPLEMENTARY MATERIALS

S1. Psychometric fitting and exclusion criteria

S1.1 Fitting procedure

Performance was modeled for each participant and condition using a cumulative Gaussian psychometric function (psignifit 4.0; Schütt et al., 2016). The function relates the signed orientation difference between test and sample (in degrees; test minus sample) to the probability of a "clockwise" response. Four parameters were estimated: mean (μ, constrained to 0), slope (σ), lower asymptote (γ; guess rate), and upper asymptote (λ; lapse rate). Parameter constraints: $\gamma \in [0, 0.5]$, $\lambda \in [0, 0.4]$, and $\gamma + \lambda < 1$. Slope estimates were required to be positive ($\sigma > 0$), and derived precision ($1/\sigma$) was required to be less than 1000 to exclude degenerate near-step fits in which σ collapses to near-zero. Fits were computed separately for each participant in each condition.

S1.2 Exclusion criteria

Participants were included if all condition-level fits yielded valid, bounded parameter estimates satisfying all constraints described above. This required valid fits in all conditions (Experiment 1: 2 mask conditions; Experiment 2: 3 loads X 2 mask conditions; Experiment 3: 5 mask durations + no mask). Participants whose fits violated any constraint for a given condition were excluded from all analyses.

S2. Effective precision analyses

In cumulative Gaussian models with free lower and upper asymptote parameters, the slope ($1/\sigma$) and asymptote (γ, λ) can trade off: when the psychometric function collapses toward chance (e.g., as it does on a subset of masked iconic trials in Experiment 1), σ becomes weakly identified and may spuriously inflate the raw precision estimate. That is, when guess rates are high, the informative portion of the psychometric function occupies only a small fraction of the response space, and maximum-likelihood fitting may assign a steep slope to that narrow range, producing an artificially high $1/\sigma$ that does not reflect genuine representational fidelity (Wichmann & Hill, 2001; Prins, 2012). We therefore computed an effective precision measure that scales the raw slope by the estimated proportion of trials containing usable information:

$$\text{effective precision} = (1 - \gamma - \lambda) \cdot (1/\sigma)$$

where γ is the lower asymptote (guess rate) and λ is the upper asymptote (lapse rate). This quantity corresponds to the slope of the observed psychometric function at its midpoint— a standard measure of psychometric steepness in the presence of nonzero asymptotes, distinct from the slope of the latent sensory function $1/\sigma$ (Strasburger, 2001). This quantity captures the fidelity of the representation conditional on information being available at all: when guess rates are zero and lapse rates are minimal, effective precision reduces to raw $1/\sigma$. When larger subsets of trials are 'lost' to guessing, effective precision is discounted accordingly. By reporting adjusted slope estimates weighted by the dynamic range of the psychometric function alongside standard precision and guess parameters, this measure resolves the discrepancy of the masking manipulation appearing to increase precision when guess rates are high (as in Experiment 1).

S3. Experiment 3: Iconic disruption does not scale with mask duration

Experiment 1 established that dynamic, feature-matched masks reliably elevate guess rates during iconic retention while leaving the precision of surviving representations intact. Before applying the masking manipulation to a multi-second working memory delay, Experiment 3 characterized the temporal boundary of vulnerability to masking by parametrically varying mask exposure duration within the iconic window tested in Experiment 1. If, as the results from Experiment 1 suggest, iconic memory depends on an overlap-vulnerable sensory-format trace, then only masks whose onset coincides with sample offset should reliably disrupt performance, and masks that appear later in the retention interval, after the trace has decayed, should not. Experiment 3 tested this directly by anchoring mask offset to probe onset and varying mask onset across five durations (50–400 ms).

S3.1 Methods

The spatial layout, stimulus properties, and mask characteristics of Experiment 3 were identical to those used in Experiment 1. As in Experiment 1, each trial consisted of a 50 ms sample grating followed by a fixed 400 ms retention interval; at test, participants reported whether the probe grating was rotated clockwise or counterclockwise relative to the sample. On mask trials, the sample location was flooded with dynamic, feature-matched gratings that always terminated at probe onset. Mask duration was varied across five levels: 50, 100, 200, 300, and 400 ms. Because mask offset was anchored to probe onset, mask onset time varied as a function of duration (onset = 400 − duration ms). Longer durations therefore produced earlier mask onset and more sustained occupation of the same feature-selective channels during retention. No-mask trials were identical except the interval remained blank. 40 participants were recruited; 12 were excluded due to at least one invalid psychometric fit (using the same exclusion criteria detailed in **Supplementary S1**), leaving N = 28 in all subsequent analyses.

S3.2 Results

S3.2.1 Precision

Raw precision (1/σ) did not reliably differ between no-mask and masked trials when collapsed across mask durations: M(no mask) = 0.29, M(mask) = 0.19, paired t(27) = −1.13, p = .27, BF_{10} = 0.36, favoring the null. A linear mixed-effects model on masked trials only showed no reliable trend across mask durations (slope = .00019/ms, t(138) = 0.74, p = .46, BF_{10} = 0.11).

S3.2.2 Guess rate

Collapsed across durations, masked trials produced significantly higher guess rates than no-mask trials (t(27) = 3.62, p = .001, BF_{10} = 48.11), providing strong evidence for an effect of masking. A linear mixed-effects model on masked trials revealed a significant positive trend, with mask duration slope = .000418/ms; t(138) = 6.61, p < .001, $BF_{10} > 10^6$.

Follow-up analyses indicated that this apparent linear trend was driven almost entirely by the 400 ms condition (the only condition in which mask onset coincided with sample offset.) A one-way repeated-measures ANOVA across all six conditions confirmed that at least one condition differed: F(5, 135) = 20.56, p < .001, $BF_{10} > 10^{14}$. Bonferroni-corrected pairwise contrasts against the no-mask baseline (corrected α = .01) showed that the 50, 100, 200, and 300 ms conditions did not reliably differ from no-mask (all

$|t(27)| \leq 1.74$, all $p \geq .09$, all $BF_{10} \leq 0.84$), whereas the 400 ms condition produced a large increase: $t(27) = 6.16$, $p < .001$, $BF_{10} = 4.07 \times 10^4$. A planned contrast comparing 400 ms against the pooled 50–300 ms conditions yielded a similarly large effect: $t(27) = 5.96$, $p < .001$, $BF_{10} = .00024$. A sensitivity analysis refitting the duration trend after excluding the 400 ms condition eliminated the linear effect (slope = .000073/ms, $t(110) = 1.18$, $p = .24$, $BF_{10} = 0.19$).

S3.2.3 Effective precision

Effective precision revealed a non-significant reduction under masking (collapsed across mask durations: $t(27) = -1.98$, $p = .059$, $BF_{10} = 1.25$). A linear mixed-effects model on masked trials showed no reliable trend with mask duration (slope = −.00014/ms, $t(138) = -1.57$, $p > .1$, $BF_{10} = 0.29$).

Together, these results suggest that the masking effect at 400 ms is concentrated in the guessing parameter: when masks coincide with sample offset, a larger proportion of trials lose access to the iconic trace entirely, while the precision of any surviving representation remains unaffected.

S4. Model-free analyses

To 1) confirm that the iconic masking effect was detectable using a coarse accuracy metric and 2) provide a measure independent of psychometric model assumptions, we examined performance on the smallest orientation difference (±10°), scoring responses as correct or incorrect by sign (excluding any 0° trials).

S4.1 Experiment 1

Accuracy for the smallest orientation difference (±10°) was lower in mask trials than no-mask trials ($t(27) = -2.87$, $p = .008$, $BF_{10} = 7.91$), providing convergent evidence with the psychometric modeling results for Experiment 1.

S4.2 Experiment 2

Accuracy for the ±10° orientation difference, collapsed across load, did not differ reliably between mask and no-mask trials ($t(28) = -1.65$, $p = .11$, $BF_{10} = 0.72$), again converging with the psychometric finding that masking did not reliably disrupt performance in the working memory task.

S4.3 Experiment 3

Accuracy for the ±10° orientation difference, collapsed across mask durations, was lower in mask trials than no-mask trials, $t(27) = -7.40$, $p < .001$, $BF_{10} > 10^6$. Within masked trials, accuracy declined with mask duration (slope = −.00048/ms, $t(138) = -6.91$, $p < .001$, $BF_{10} > 10^7$), and was again lowest in the 400 ms condition.

These analyses converge with the psychometric modeling results: masking selectively impaired iconic performance by eliminating access to orientation information on a substantial proportion of trials, reducing accuracy to near-chance. The agreement between the model-based and model-free analyses supports the reliability of the psychometric data and provides further evidence that the observed masking effects in Experiments 1 and 3 are not artifacts of the fitting procedure.

S5 Cross-experiment comparison (Iconic vs. VWM)

To formally test whether the masking manipulation differentially affected iconic persistence and visual working memory, we compared the mask effect on guess rate (γ) across Experiments 1 and 2 using a 2 (experiment: Iconic, VWM; between-subjects) × 2 (mask: no-mask, mask; within-subjects) linear mixed-effects model with a random intercept by participant. The experiment × mask interaction — the formal test of the cross-experiment dissociation — was decisive, $F(1, 57) = 18.24$, $p < .001$, $BF_{10} = 256$. Masking elevated guess rate by 0.112 in Iconic (M(no mask) = 0.051, M(mask) = 0.162) but only by 0.017 in VWM, collapsed across load (M(no mask) = 0.058, M(mask) = 0.075). The same masking condition that substantially disrupted iconic retention left visual working memory largely unaffected. This dissociation was further corroborated by an independent-samples t-test directly comparing per-subject mask-effect magnitudes (no mask − mask) between experiments, $t(55) = -4.20$, $p < .001$, $BF_{10} = 362$.